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CLINICAL ARTICLE

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Pure arterial malformations

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OBJECTIVE Over the last half century, there have been isolated case reports of purely arterial malformations. In this study, the authors report a consecutive series of patients with pure arterial malformations, emphasizing the clinical and radiological features of these lesions.

METHODS Pure arterial malformations were deÿned as dilated, overlapping, and tortuous arteries with a coil-like appearance and/or a mass of arterial loops without any associated venous component. Demographic characteristics of the patients, cardiovascular risk factors, presentation, radiological characteristics, and follow-up data were collected. Primary outcomes were new neurological symptoms including disability, stroke, and hemorrhage.

RESULTS Twelve patients meeting the criteria were identiÿed. Ten patients were female (83.3%) and 2 were male (16.6%). Their mean age at diagnosis was 26.2 Ó 11.6 years. The most common imaging indication was headache (7 patients [58.3%]). In 3 cases the pure arterial malformation involved the anterior cerebral arteries (25.0%); in 4 cases the posterior communicating artery/posterior cerebral artery (33.3%); in 2 cases the middle cerebral artery (16.6%); and in 1 case each, the superior cerebellar artery, basilar artery/anterior inferior cerebellar artery, and posterior inferior cerebellar artery. The mean maximum diameter of the malformations was 7.2 Ó 5.0 mm (range 3ñ16 mm). Four lesions had focal aneurysms associated with the pure arterial malformation, and 5 were partially calciÿed. In no cases was there associated intracranial hemorrhage or infarction. One patient underwent treatment for the pure arterial malformation. All 12 patients had follow-up (mean 29 months, median 19 months), and there were no cases of disability, stroke, or hemorrhage.

CONCLUSIONS Pure arterial malformations are rare lesions that are often detected incidentally and probably have a benign natural history. These lesions can affect any of the intracranial arteries and are likely best managed conservatively.

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KEY WORDS vascular malformation; aneurysm; vascular disorders

Over the last half century, there have been isolated case reports of purely arterial malformations deþned as dilated, overlapping, and tortuous arteries with a coil-like appearance and/or a mass of arterial loops without any associated venous component.[1–3,5–7,9–11,15–21,23–31] Now deþned as òpure arterial malformations,ó these lesions are commonly mistaken for aneurysms or arteriovenous malformations on noninvasive imaging.[16,17,19,20] Unlike aneurysms and arteriovenous malformations, they are thought to have a benign natural history and are generally managed conservatively. Herein we report on a consecutive series of 12 patients with pure arterial malformations, with an emphasis on the clinical and imaging manifestations of these lesions.

Methods

Patient Selection and Disease DeĀnition

Following IRB approval, we searched the senior author's intracranial aneurysm and vascular malformation database for cases of pure arterial malformation, dissecting aneurysm, and dissection and contacted a second author to identify any additional cases from his institution. We also searched our institutionõs imaging database for reports indicating vessel dilation or dolichoectasia. In deþning pure arterial malformations, we used the deþnition put forth by McLaughlin et al.: dilated, overlapping, and tortuous arteries forming a mass of arterial loops with a coil-like appearance in the absence of any arteriovenous

ABBREVIATIONS ACA = anterior cerebral artery; AICA = anterior inferior cerebellar artery; BA = basilar artery; ICA = internal carotid artery; MCA = middle cerebral artery; PCA = posterior cerebral artery; PCoA = posterior communicating artery; PICA = posterior inferior cerebellar artery; SCA = superior cerebellar artery.

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TABLE 1. Description of cases in our single-institution series of patients with pure arterial malformations

Case No.	Age (yrs), Sex	Reason for Imaging	Lesion Location	Imaging Appearance	Max Diameter (mm)	Associated Symptoms	Imaging FU	Clinical FU
1	10, F	Severe lt-sided HAs	Lt supraclinoid ICA, PCoA, & PCA	Multilobulated pseudoaneurysm of supraclinoid ICA, partially calciÿed, coil-like tortuosity & dilatation of PCoA & PCA, stenosis of left M_1	13	Yes	72 mos, no change	72 mos, no change
2	19, F	Incidental	Lt MCA	Coil-like tortuosity of distal lt M_1 & lenticulostriate vessel w/ superimposed multilobulated aneurysm, mild preceding stenosis	16	No	NA	36 mos, no change
3	27, F	HA, lt hemibody numbness, & facial droop	BA	Tortuous BA, tightly wound, mildly dilated, no focal aneurysmal dilatation	5	Possibly	NA	5 mos, no change
4	25, F	HA	Lt supraclinoid ICA & M_1	Tortuous & coil-like appearance of supraclinoid ICA & M_1 w/ 3 focal aneurysms, calciÿed	8	No	60 mos, no change	60 mos, no change
5	25, F	HA after minor trauma	Lt ACA	Tortuous A_2, mildly dilated	4	No	1 mo, no change	30 mos, no change
6	34, F	Light trauma	Lt ACA	Tortuous A_2, mildly dilated, calciÿed	4	No	3 mos, no change	12 mos, no change
7	38, F	Transient hand numbness	Lt PICA	Tortuous, tightly coiled	3	No	NA	2 mos, no change
8	11, M	Incidental	Lt PCoA	Tortuous & coil-like appearance of supraclinoid ICA & PCoA	5	No	NA	1 mo, no change
9	17, M	HA	Rt SCA	Tortuous SCA, tightly wound, moderately dilated, no focal aneurysmal outpouching	4	No	NA	26 mos, no change
10	47, F	Prior thunderclap HA	Rt ACA	Tortuous artery, mildly dilated, partially calciÿed, delayed venous drainage	4	No	1 mo, no change	27 mos, no change
11	35, F	HA	Lt PCoA/PCA	Arterial tortuosity w/ focal aneurysm dilatations, calciÿed	15	No	84 mos, no change	84 mos, no change
12	20, F	Trauma	Rt ICA, P_2 & lt P_2	Tortuous, tightly coiled	3	No	NA	1 mo, no change

FU = follow-up; HA = headache; NA = not available.

connection.[20] Only cases that met this deþnition as agreed upon by the study's authors were included.

Imaging Analysis

For each patient, we performed a comprehensive evaluation of all associated brain imaging studies. CT and MR images were evaluated for the presence of acute or chronic infarction; white matter disease; vascular calciþcations; T1 signal abnormality of the vessel wall indicating presence of dissection; malformations of cortical development; and abnormal enhancement. Cross-sectional and conventional angiographic images were evaluated for evidence of aneurysms or stenosis associated with the pure arterial malformation, maximum diameter, location, and interval growth if follow-up imaging was available.

Clinical Chart Review

All patients received a follow-up telephone call from a research nurse to determine if the patients had suffered any disability, stroke, or intracranial hemorrhage since they last presented to our institutions. Modiþed Rankin Scale scores were collected for all patients. Follow-up was performed 1 month prior to submission of this article. Clinical charts were reviewed to determine the indication for the imaging examination in which the pure arterial malformation was discovered, any history of trauma, any association between the vascular abnormality and symptoms, and any new neurological symptoms including infarction, transient ischemic attack, or hemorrhage associated with the lesion at follow-up. Baseline characteristics including demographic variables and history of cardiovascular risk factors such as smoking, hypertension, hyperlipidemia, coronary artery disease, and diabetes mellitus were collected. We also collected data on any personal or family history of connective tissue disease and aneurysm.

Statistical Analysis

This study is a descriptive study. No statistical comparisons were made. Continuous variables are reported as the mean Ñ standard deviation and range, and categorical variables are reported as number (%). Data analysis was performed using the SAS-based statistical software package JMP version 12.0 (www.jmp.com).